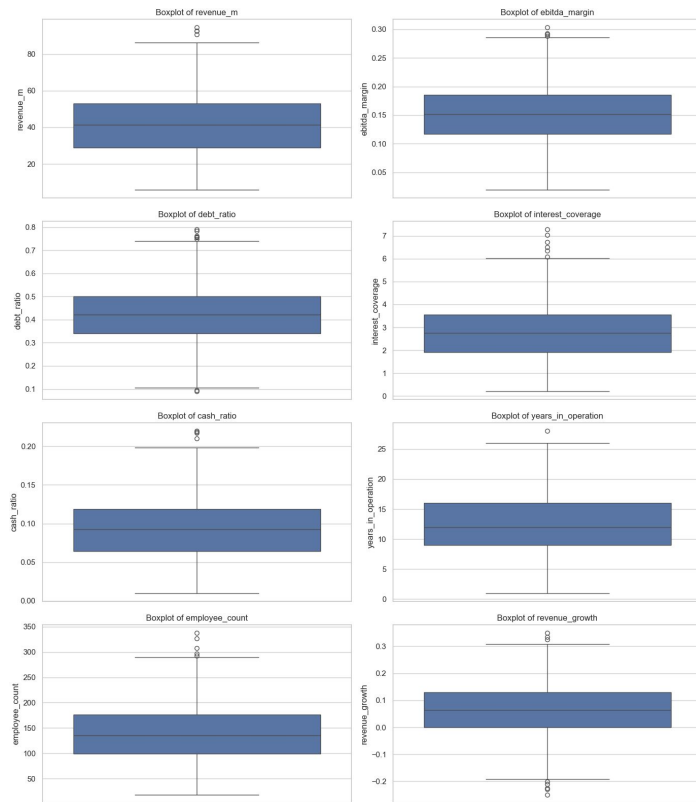
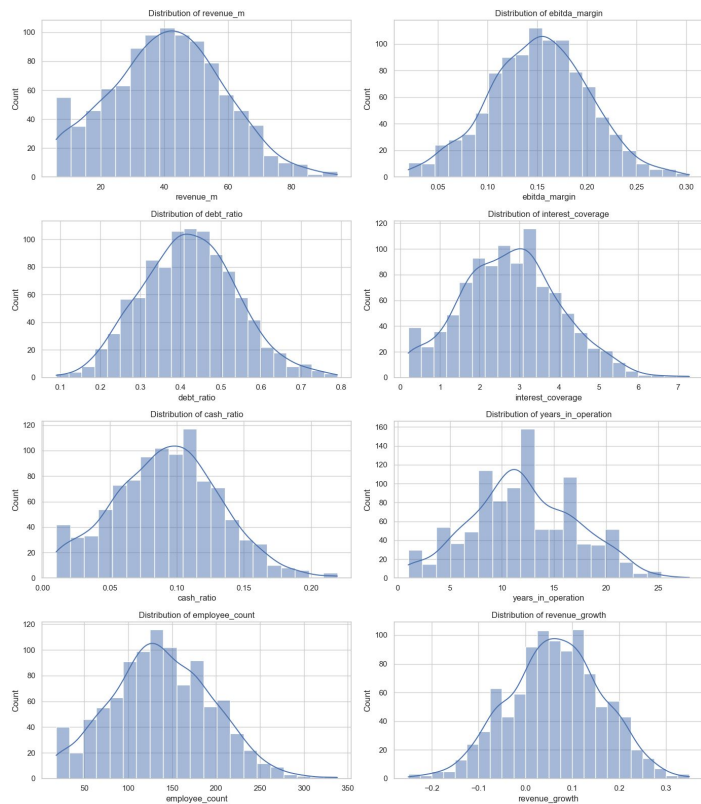


Fasanara Capital Assessment

Enea Pedretti

Data: class imbalance



Feature Engineering

- New features: Debt coverage profitability, debt to earning, distress score...
- 1 hot encoding for not-continuous features
- Every description is assembled from a small set of standard phrases: "recurring subscription revenue", "declining revenue", "regulatory uncertainty", and so on. No embedding is needed. Identify strength (1) and weakness (2) phrase patterns.

Analyst-like explanations

- Template prompt with 3 largest positive and negative SHAP features
- Raw narrative text

Modeling

- Base line: logistic regression with balanced class weight
- Main model: XGBoost with hyperparameters search + platt scaling + threshold tuning
- See Streamlit

Model	Threshold	ROC-AUC	PR-AUC	Brier Score	ECE	pMAD	ECE/pMAD Ratio	Recall (TPR)	FPR	FNR
Logistic Regression	0.5000	0.7603	0.4503	0.1862	21.32%	22.93%	0.9296	64.94%	25.91%	35.06%
Tuned XGBoost	0.5000	0.7790	0.4836	0.1570	16.13%	21.76%	0.7412	63.22%	20.46%	36.78%
Calibrated XGBoost (Platt)	0.0982	0.7797	0.4712	0.1192	2.62%	11.15%	0.2349	86.78%	49.76%	13.22%

Samuele Mazzanti:

<https://medium.com/data-science-collective/it-took-me-6-years-to-find-the-best-metric-for-classification-models-0f5aa21a2b85>